Agenda

- Introduction
- OOP Concepts
- Hello world Program
- Data Types
- Console input and output

Java History

- In 1991, group of sun engineers led by James Gosling and Patrick Naughton decided to design a language that could run on small devices like remote controls, cable tv boxes.
- Since these devices have very small power and memory the language needs to be small.
- Also diffent manufactures can choose different CPU's the language cannot be bound to single architecture
- this project was named as green.
- these engineers came from UNIX background, so they used c++ as their base.
- James decided to call this language as OAK, however the language with this name was already existing, Hence it was later renamed by James to Java.
- In 1992 they delivered their first product called as "*7"(a smart remote control)
- Unfortunately Sun Miccrosystem was not intrested in producing this, also nor the consumer electronic companies were intrested in it.
- The team then deciced to market their technology in some other way where they worked for next 1 and half year on it.
- Meanwhile world wide web (www) was growing bigger.
- the key to it was browser transalating hyper text pages to the screen.
- the java developers developed a browser called as HotJava browser which was based on client server architeture and was working in real time.
- the developers made the browser capable of executing java code inside the web pages called as Applets.

Java Versions

- JDK Beta - 1995
- JDK 1.0 - January 23, 1996
- JDK 1.1 - February 19, 1997
- J2SE 1.2 - December 8, 1998
  - Java collections
- J2SE 1.3 - May 8, 2000
- J2SE 1.4 - February 6, 2002
- J2SE 5.0 - September 30, 2004
  - enum
  - Generics
  - Annotations
- Java SE 6 - December 11, 2006
- Java SE 7 - July 28, 2011
- Java SE 8 (LTS) - March 18, 2014

    - Functional programming: Streams, Lambda expressions
- Java SE 9 - September 21, 2017
- Java SE 10 - March 20, 2018
- Java SE 11 (LTS) - September 25, 2018
- Java SE 12 - March 19, 2019
- Java SE 13 - September 17, 2019
- Java SE 14 - March 17, 2020
- Java SE 15 - September 15, 2020
- Java SE 16 - March 16, 2021
- Java SE 17 (LTS) - September 14, 2021
- Java SE 18 - March 22, 2022
- Java SE 19 - September 20, 2022
- Java SE 20 - March 21, 2023

Java Platforms

- Java is not specific to any processor or operating system as it is implemented for wide variety of hardware and operating system
- 1. Java Card
    - used to run java based applications on small devices with small memory devices like smart cards
- 2. Java ME(Micro Edition)
    - used to develop applications for small devices with less memory, display and power capacities like mobiles, printers
- 3. Java SE(Standard Edition)
    - It is widely used for development of portable code for desktop environment
- 4. Java EE(Enterprise Edition)
    - It is widely used in development of enterprise applications/softwares. -also used for web application development

Java Installation

- Windows and Mac:
- Download .msi/.dmg file and follow installation steps.

```
https://adoptium.net/temurin/releases/?version=11
```

Eclipse STS 3.9.18

- Link

```
https://github.com/spring-attic/toolsuite-distribution/wiki/Spring-Tool-Suite-3
```

JDK vs JRE vs JVM

- SDK -> Software Development Kit

- SDK = Software Development Tools + Libraries + Runtime environment + Documentation + IDE
  - Software Development Tools = Compiler, Debugger, etc.
  - Libraries = Set of functions/classes.
- JDK -> Java Development kit
  - used for developing Java applications.
- JDK = Java Development tools + Java docs + JRE
- JRE = Java API(Java class libraries) (rt.jar replaced by jmods in java9) + Java Virtual Machine
  - till java 8
    - JRE = rt.jar + JVM
  - from java 9
    - JRE = jmods + libraries + JVM

Documentation and tutorial Link

```
- Java SE 8 Document Link
https://docs.oracle.com/javase/8/docs/api/index.html
```

Object Oriented

- basic principles of OOP are
- 1. class
- 2. object

class

- It is a logical entity
- It is a user defined datatype (same as struct in c)
- It consists of field(data members) and methods(member functions)
- Methods
  - static methods -> Accessed using classname directly
  - non static methods-> Accessed using object of the class
- It is also called as blueprint of object/instance

Object

- It is a physical entity
- It is an instance of a class
- one class can have multiple objects
- Object is created in java using new operator

HelloWorld

```
class Program{
    public static void main(String args[]){
        System.out.println("Hello World");
```

```
    }
}
```

main()

- In java every variable/function should be class (Encapsulation)
- JVM calls main method without creating object of the class, so main method should be static
- main does not return anythinig to JVM so it is void.
- main takes command line arguments and hence String args[]
- main should be accessiable outside the class directly and hence public.
- JVM invokes main method.
- Can be overloaded.
- Can write one entry-point in each Java class.

System.out.println()

- System is predefined Java class (java.lang.System).
- out is public static field of the System class --> System.out.
- out is object of PrintStream class (java.io.PrintStream).
- println() is public non-static method of PrintStream class

Public class

- As per Java Langauage Specification
  - 1. Name of public class and name of java file should be same.
  - 2. A single .java file can have only 1 public class.
  - 3. A single .java file can have multiple non public classes.

DataTypes

- It defines 3 things
  - 1. Nature (type of data stored)
  - 2. Memory (Memory required to store the data)
  - 3. Operations (what operations we can perform)
- Java is Strictly type checked language
- In java, data types are classified as:

1. Primitive types or Value types
2. Non-primitive types or Reference types

```
Data types
    |- Primitive types (Value types)
    |       |- Boolean: boolean
    |       |- Character: char
    |       |- Integral: byte, short, int, long
    |       |- Floating-point: float, double
    |
    |- Non-Primitive types (Reference types)
```

```
            |- class
            |- interface
            |- enum
            |- Array
```

Datatype	Detail	Default	Memory needed (size)	Examples	Range of Values
boolean	It can have value true or false, used for condition and as a flag.	false	1 bit	true, false	true or false
byte	Set of 8 bits data	0	8 bits	NA	-128 to 127
char	Used to represent chars	\u0000	16 bits	"a", "b", "c", "A" and etc.	Represents 0-256 ASCII chars
short	Short integer	0	16 bits	NA	-32768-32768
int	integer	0	32 bits	0, 1, 2, 3, -1, -2, -3	-2147483648 to 2147483647-
long	Long integer	0	64 bits	1L, 2L, 3L, -1L, -2L, -3L	-9223372036854775807 to 9223372036854775807
float	IEEE 754 floats	0.0	32 bits	1.23f, -1.23f	Upto 7 decimal
double	IEEE 754 floats	0.0	64 bits	1.23d, -1.23d	Upto 16 decimal

Console input and output

- Java has several ways to take input and print output. Most popular ways in Java 8 are
- 1. using Scanner class (Program01)
    - it is present in java.util package

```
Scanner sc = new Scanner(System.in);
System.out.print("Enter name: ");
String name = sc.nextLine();
System.out.print("Enter age: ");
int age = sc.nextInt();
System.out.println("Name: " + name + ", Age: " + age);
System.out.printf("Name: %s, Age: %s\n", name, age);
```